CSA5108 Cryptography and Network security

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Exp 14 : One-Time-Pad Version of the Vigenère Cipher
Code:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main(void) {
   char p[200], c[200], target[200];
   int key[200], n=0, k, m=0;

   printf("Enter plaintext: ");
   fgets(p,sizeof(p),stdin);

   for(int i=0;p[i];i++)
      if(isalpha((unsigned char)p[i]))
         p[n++]=tolower((unsigned char)p[i]);
   p[n]='\0';

   printf("Enter %d key numbers (0-25): ",n);
   for(int i=0;i<n;i++) {
      if(scanf("%d",&key[i])!=1 || key[i]<0 || key[i]>25) {
         printf("Invalid key.\n");
         return 1;
      }
   }

   for(int i=0;i<n;i++)
      c[i]=(p[i]-'a'+key[i])%26+'a';
   c[n]='\0';

   printf("Ciphertext: %s\n",c);
```

```
    printf("Enter target plaintext: ");
    scanf("%199s",target);

    if(strlen(target)!=strlen(c)) {
        printf("Target plaintext must have the same length.\n");
        return 1;
    }

    printf("New key stream: ");
    for(int i=0;i<n;i++) {
        k=(c[i]-'a'-(tolower((unsigned char)target[i])-'a')+26)%26;
        printf("%d ",k);
    }

    printf("\n");
    return 0;
}
```

